

PMaaS 4.0 × i3 Agentic AI Platform

AI Integration Blueprint · Open-Source Model Expansion · Interface Design Advisory

PMaaS 4.0

i3 Platform

KIMI K3 · GLM · Qwen

Co-Worker Interface

IBM watsonx

Kenya 2027

Part 1 — Understanding What PMaaS 4.0 Is

PMaaS 4.0 is a **full-cycle AI-powered political operating system** for Kenya's 2027 General Elections. After reading the complete strategy document, here are the 8 core AI technical components it requires:

#	PMAAS COMPONENT	WHAT IT DOES	TECH CURRENTLY SPECIFIED
1	AI Voice-Clone Contact Centre	Candidate's voice calls every voter 24/7 in Swahili/English. Handles Q&A on manifesto.	ElevenLabs + IBM Watson TTS + Twilio / Africa's Talking
2	Agentic Policy Knowledge Bot	RAG over full manifesto, speeches, policy papers. Answers any voter question.	watsonx Orchestrate + watsonx.ai RAG
3	Real-Time Sentiment Heatmap	Ward-level live political intelligence from 7 data streams.	IBM Granite + NLP + custom dashboards
4	Deepfake Defense + Truth Engine	Monitors synthetic media attacks, deploys watermarked response in 90 min.	Custom AI monitoring
5	WhatsApp AI Bot	Text-based policy advisor, registration guide, donation flows, gamified quiz.	WhatsApp Business API + watsonx
6	Social Virality Engine	AI-generated TikTok/Reels scripts, trend-scan every 2 hours, influencer matching.	watsonx generative AI
7	Campaign War Room Intelligence	6 autonomous AI agents: Sentiment Alert, Content Scheduling, Volunteer Coord., Budget Burn, Misinformation Hunter, Sales Follow-up.	watsonx Orchestrate multi-agent
8	Governance AI (Post-Election)	Manifesto delivery tracker, constituency feedback bot, Hansard monitor, 'Ask Your MP' bot.	Persistent RAG + CRM integration

Part 2 — How PMaaS Fits Into the i3 Agentic AI Platform

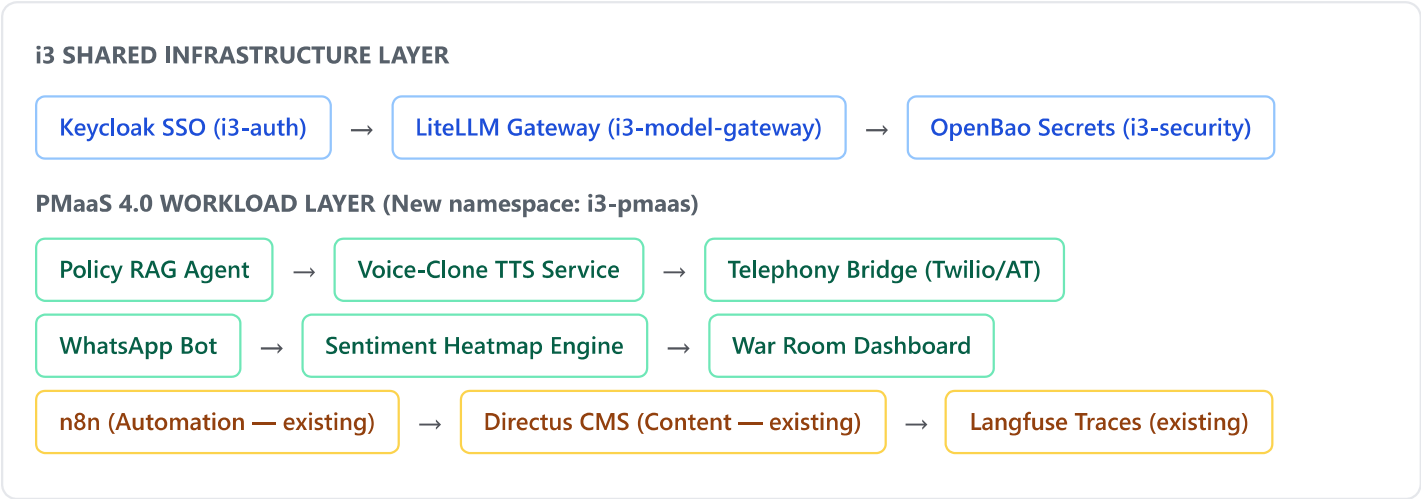
The Integration Is Natural — Not a New Build

The i3 platform already has LiteLLM (model gateway), RAG (ChromaDB + LlamaIndex), FastAPI agents, n8n automation, Directus CMS, OTT streaming, Keycloak SSO, and Langfuse tracing. PMaaS 4.0 needs ALL of these — it becomes a

vertical deployment

on the same infrastructure, not a separate system.

Proposed Architecture: PMaaS as a Namespace on i3 Platform



What Each Existing i3 Component Provides to PMaaS

i3 COMPONENT	PMAAS USE CASE	INTEGRATION EFFORT
LiteLLM Model Gateway	Routes all PMaaS LLM calls: policy Q&A, sentiment, content generation, translation (Swahili/Sheng)	Low — add PMaaS API keys + model aliases
ChromaDB + LlamaIndex RAG (Admissions pattern)	Ingest full manifesto, speeches, ward data. Policy Knowledge Bot answers from this. Clone the admissions_agent.py pattern.	Low — clone & adapt existing agent
n8n Automation	Voter registration flows, polling day call triggers, WhatsApp broadcast scheduling, Nyumba Kumi task dispatch	Medium — build PMaaS-specific workflows
Directus CMS	Manifesto content management, ward-specific message variants, social content calendar, merchandise QR pages	Low — new collections in existing CMS
OTT Stack (OME + nginx-hls)	Livestream campaign rallies, candidate video messages, VOD recordings with AI-generated subtitles	None — already deployed
Langfuse	Trace every voter interaction: what they asked, what model answered, sentiment score, cost per call	None — automatic via LiteLLM
Keycloak SSO	War Room access control: Campaign Manager, Field Agent, Analyst, Candidate roles	Low — add PMaaS realm clients
Grafana + Prometheus	War Room operational dashboards: call volumes, sentiment trends, ward-level heatmap data	Medium — build PMaaS dashboards

New PMaaS-Specific Services to Build (i3-pmaas namespace)

1. Policy RAG Microservice

2. Voice-Clone TTS Microservice

Clone of admissions_agent.py — FastAPI + LlamaIndex + ChromaDB. Ingests manifesto, ward data, speech transcripts. Answers voter questions in Swahili/English/Sheng via voice bot and WhatsApp.

Clone admissions_agent.py

ElevenLabs API wrapper + IBM Watson TTS fallback. Converts RAG responses to candidate-voice audio. Served as WAV/MP3 stream to Twilio/Africa's Talking telephony layer.

New service — FastAPI

3. Sentiment Heatmap Engine

Ingests 7 data streams (social, SMS, IVR, community WhatsApp, media, rival, IEBC). Runs Granite/Qwen sentiment classification. Outputs ward-level scores to PostgreSQL → Grafana.

New service — Python worker

4. Deepfake Monitor Agent

Scheduled scraper of X/TikTok/Facebook for synthetic media mentioning candidate name. Uses vision model (Granite Vision) for image/video authenticity scoring. Alert + Truth Response queue.

New — n8n + vision model

5. Telephony Bridge

Twilio / Africa's Talking IVR adapter. Handles inbound/outbound call routing. Connects TTS microservice to GSM network. Streams candidate voice in real time. DTMF sentiment capture.

New — Node.js / FastAPI

6. War Room Dashboard

Next.js frontend (or Grafana extended) showing live heatmap, call volumes, sentiment by ward, rival activity alerts, budget burn, volunteer status, social content queue.

New — Next.js + Grafana

🌐 Part 3 — Adding KIMI K3, GLM & Qwen to the Model Gateway

🧠 Why These Models Matter for PMaaS + i3 Platform

KIMI K3 (Moonshot AI), GLM-4 (Zhipu AI), and Qwen 2.5 (Alibaba) are world-class open-weight models with exceptional multilingual performance — critical for Swahili, Arabic, and African language support that IBM Granite does not cover as well. All three can be added through the **existing LiteLLM gateway** with zero architectural changes.

Model Profiles — What Each Brings

MODEL	MAKER	SIZE	STRENGTHS	BEST PMAAS / I3 USE CASE
KIMI K3 Moonshot AI	China	~236B MoE (open weights)	Exceptionally long context (128K+), deep reasoning, strong at structured document understanding, multilingual	Manifesto deep-dive RAG (long doc), Hansard analysis, opposition research dossiers, legal document analysis
GLM-4 Zhipu AI	China	9B / 130B variants	Best-in-class Chinese + multilingual, strong tool-calling / function call, code, long-context understanding	Swahili/Sheng policy chat, WhatsApp bot NLU, voter sentiment classification, multi-turn dialogue

MODEL	MAKER	SIZE	STRENGTHS	BEST PMAAS / I3 USE CASE
Qwen 2.5 Alibaba	China	0.5B–72B (open weight)	Outstanding multilingual (29 languages incl. Swahili), best-in-class coding (Qwen-Coder), Math, instruction following	Swahili voter chatbot, code generation for EvalOS, AI Lab notebooks, rapid content generation in multiple languages
KIMI K3 (API) Moonshot API	China	API-only	Long document summarisation, research, agentic tool use	Tier 1 replacement alongside watsonx for complex agentic tasks

Current Model Stack + Additions

IBM Granite 3.1 2B (Tier 3)

DistilBERT Intent (Tier 3)

Mistral NeMo 12B (Tier 2)

DeepSeek-Coder 16B (Tier 2)

IBM Granite 8B watsonx (Tier 1)

Granite Vision 2B (Tier 1)

+ Qwen2.5 7B / 72B (NEW)

+ GLM-4 9B (NEW)

+ KIMI K3 API (NEW)

How to Add Them — LiteLLM Config Additions

```
# Add to litellm-deploy.yaml config.yaml → model_list:

# — Qwen 2.5 72B (via Ollama local) —————
- model_name: qwen25-72b
  litellm_params:
    model: ollama/qwen2.5:72b
    api_base: http://ollama-service.i3-model-gateway.svc.cluster.local:11434
    timeout: 90
    max_tokens: 32768
  model_info:
    tier: 2
    description: "Qwen 2.5 72B - multilingual Swahili/Sheng + coding"

# — Qwen 2.5 7B (fast edge – Swahili chat) —————
- model_name: qwen25-7b
  litellm_params:
    model: ollama/qwen2.5:7b
    api_base: http://ollama-service.i3-model-gateway.svc.cluster.local:11434
    timeout: 30
    max_tokens: 8192
  model_info:
    tier: 3
    description: "Qwen 2.5 7B - fast Swahili voter chatbot"

# — GLM-4 9B (multilingual tool calling) —————
- model_name: glm4-9b
  litellm_params:
    model: ollama/glm4:9b
```

```
api_base: http://ollama-service.i3-model-gateway.svc.cluster.local:11434
timeout: 45
max_tokens: 8192
model_info:
  tier: 2
  description: "GLM-4 9B - Swahili NLU, function calling"

# — KIMI K3 (via Moonshot API - Tier 1) —————
- model_name: kimi-k3
  litellm_params:
    model: moonshot/moonshot-v1-128k
    api_key: "os.environ/MOONSHOT_API_KEY"
    timeout: 120
    max_tokens: 65536
  model_info:
    tier: 1
    description: "KIMI K3 - long-context manifesto analysis"
```

Routing Strategy — Which Model for Which Task

TASK	RECOMMENDED MODEL	WHY
Voter policy Q&A (Swahili)	Qwen2.5-7B (fast) → GLM-4-9B (complex)	Best Swahili instruction following at low latency
Manifesto deep analysis / long docs	KIMI K3 (128K context)	Can hold entire manifesto in context — no chunking needed
Sentiment classification (ward-level)	Qwen2.5-7B or GLM-4-9B	Fast, multilingual, strong classification
Code generation (EvalOS, AI Lab)	Qwen2.5-Coder or DeepSeek-Coder	Both world-class at coding
Content generation (social media)	Qwen2.5-72B or Mistral NeMo 12B	Creative, long-form multilingual content
Governance AI / Hansard analysis	KIMI K3	Long legal documents, structured extraction
Intent routing (fast)	DistilBERT (existing) / Qwen2.5-0.5B	Sub-50ms for triage

Part 4 — AI Interface Design: All Surfaces



The i3 Interface Strategy

The platform needs 4 distinct interface surfaces, each with a different persona and purpose — from a public chatbot to a private co-worker assistant. All share the same LiteLLM backend and Keycloak authentication.

 Surface 1 — i3 AI Chat (Public Conversational Interface)

A streaming, multi-turn conversational interface. The public face of i3's AI. Used by students, admissions prospects, and voters (PMaaS).

FEATURE	DETAIL
Chat personas	Admissions Bot · Student Tutor · Policy Advisor (PMaaS) · Governance Bot (PMaaS)
Input	Text + file upload (PDF, DOCX for RAG) + voice input (browser Speech API → STT)
Output	Streaming text tokens + rich markdown + code blocks + citations (RAG source + page)
Languages	English, Swahili, Sheng (route to Qwen/GLM for non-English)
Backend	FastAPI WebSocket → LiteLLM → Qwen/GLM/Granite based on detected language
Session memory	Last 20 turns in Redis; long-term user history in PostgreSQL
Deploy	Next.js frontend at <code>chat.i3technologies.co.ke</code> (new Route)

Innovative Features to Add:

- Language auto-detect: if Swahili detected → switch to Qwen2.5-7B automatically
- Voice input: browser mic → Whisper STT → text query → voice output TTS (PMaaS voice clone)
- Reasoning trace toggle: user can click "Show thinking" to see chain-of-thought steps
- Persona switcher: user selects "Tutor / Policy Advisor / Code Assistant" — adjusts system prompt + model
- Source citations: every RAG answer shows document title, page, and excerpt inline

 Surface 2 — i3 AI Coding Assistant (Developer Interface)

A VS Code / JupyterHub integrated coding assistant. Serves AI Lab students, platform developers, and EvalOS exam support.

FEATURE	DETAIL
Interface	VS Code extension (Language Server Protocol) + JupyterHub cell magic (<code>%ai</code>)
Primary model	Qwen2.5-Coder-32B (best open-source coding model) + DeepSeek-Coder-V2 as fallback
Capabilities	Code completion, explain code, refactor, generate tests, debug stack traces, generate docstrings
Context	Sends open file + visible code range + error message to model; returns inline diff or explanation
EvalOS integration	Students click "Hint" in EvalOS → sends current code + failing test to Qwen-Coder → returns hint (not solution)
Security	Code never leaves the cluster — Qwen/DeepSeek run on-prem via Ollama
API endpoint	<code>POST /v1/code/complete</code> on LiteLLM — maps to <code>qwen25-coder</code> alias

Innovative Features:

- Smart hint system: 3 levels — conceptual hint → pseudocode → worked example (each costs 1 "hint credit")
- Code review agent: submit PR / notebook → AI reviews for bugs, style, and efficiency
- AI-generated test cases: given a function signature → generate 10 edge-case tests automatically
- Plagiarism + MOSS integration: check student code similarity via EvalOS MOSS connector

 Surface 3 — i3 Co-Worker (Office & Business Productivity — Like Claude for Work)

The most ambitious interface. A persistent AI work assistant deeply integrated into daily business operations — email, documents, meetings, CRM, scheduling. Think **Claude for Work + Microsoft Copilot**, but running entirely on your own infrastructure with your own models.

MODULE	WHAT IT DOES	MODEL / CONNECTOR
Document Assistant	Upload any Word/PDF/Excel → AI summarises, extracts key data, answers questions, rewrites sections, generates new documents from templates	KIMI K3 (long context) + Directus CMS storage
Email & Comms Drafter	Draft professional emails, proposals, letters based on brief. Tone: formal / friendly / persuasive. Auto-detect language and respond in same language.	Qwen2.5-72B or Granite 8B
Meeting Intelligence	Record meeting audio → Whisper STT → AI summarises, extracts action items, assigns owners, creates follow-up tasks in n8n. Sends summary to attendees.	Whisper + Granite/Qwen + n8n
CRM Intelligence	Natural language queries of Odoo/Directus: "Show me all leads from Nairobi this month". AI updates records, writes call notes, suggests follow-up actions.	GLM-4 (function calling) + Odoo MCP
Presentation Builder	Brief → AI generates structured slide outline, writes speaker notes, suggests visuals. Export to PPTX template. PMaaS: auto-generate ward-specific campaign briefs.	Qwen2.5-72B + python-pptx
Research & Intelligence	Ask any research question → AI searches internal knowledge base + web → synthesises structured briefing note with citations. PMaaS: daily rival intelligence brief.	KIMI K3 + RAG + Tavily/Brave Search API
Task & Calendar AI	Natural language task creation: "Remind me to call Philip on Friday" → creates n8n task, Google Calendar event, sends WhatsApp reminder. Context-aware scheduling.	GLM-4 tool-calling + Google Calendar MCP
Financial Briefings	Upload budget spreadsheet → AI explains variances, flags overspends, generates executive summary. PMaaS: campaign budget burn vs. target by ward.	Qwen2.5-72B + pandas code agent

Co-Worker Interface Design:

- Single pane sidebar: always-on panel accessible from any app (like Copilot sidebar in Office 365)
- Context awareness: Co-Worker knows what page/app you are on — if in Directus CMS, offers "help write this content"; if in Grafana, offers "explain this spike"

- Memory: remembers your preferences, writing style, common queries, and past work within your session
- Action confirmation: before writing to any external system (CRM, calendar, email send), shows a confirmation card
- never acts autonomously on sensitive operations
- Audit trail: every Co-Worker action logged to Langfuse — who asked what, what model answered, what action was taken
- Deploy: `coworker.i3technologies.co.ke` — React app + persistent WebSocket

 Surface 4 — PMaaS War Room Interface (Campaign Operations)

A mission-control dashboard exclusively for campaign managers and field coordinators. Real-time intelligence + AI action centre in one screen.

PANEL	CONTENTS
Live Heatmap	Kenya ward map, colour-coded by support level, updated hourly. Click any ward for drill-down: top issues, sentiment trend, last call volumes, community agent report.
AI Agents Status	6 autonomous agents shown as cards: last action, next scheduled action, alert count. One-click approve/reject pending actions.
Call Centre Ops	Live call volume, active calls, sentiment breakdown (Supporter / Undecided / Opposition), escalation queue. AI-suggested calling priority for next hour.
Content Queue	AI-generated social content waiting for human approval before posting. TikTok scripts, WhatsApp broadcasts, press response drafts.
Rival Monitor	Last 24h rival activity feed. AI-generated counter-narrative drafts. Deepfake alert panel.
Volunteer Tracker	500+ volunteer status. Task completion rates. Leaderboard. AI-suggested reactivation messages for inactive volunteers.

Innovative PMaaS War Room Features:

- "Ask the War Room" chat: type "Which 10 wards need a call surge today?" → AI analyses heatmap + pledge data and responds in natural language with a prioritised list
- Closing Argument Engine: 7 days before polling — auto-generates personalised 90-second audio messages per undecided voter in cloned candidate voice
- Swing Ward Rescue Protocol: AI one-click triggers surge deployment to the 50 most winnable borderline wards
- Polling Day Dashboard: live turnout by station, pledge-vs-voted gap, transport coordination queue

 Part 5 — Innovative Ideas to Consider

 Universal Voice Layer

Every i3 interface gets a voice toggle. Chat → speak your question, hear the answer. Co-Worker → dictate

 African Language Expansion

Qwen2.5 supports 29 languages. Add Kikuyu, Luo, Kamba, Kalenjin detection to the routing layer. For

documents. EvalOS → read questions aloud. PMaaS → full TTS on all messages. Backend: Whisper STT (on-cluster) + ElevenLabs/Watson TTS out.

PMaaS this is a massive differentiator — candidate speaks to voters in their mother tongue. First in Kenya.

 AI Agent Marketplace

Let instructors and admins create custom AI agents via a no-code UI (similar to OpenAI GPTs). Define system prompt, attach documents, choose model. Deploy as a named agent on the platform. Build agent economy inside i3.

 WhatsApp as Universal Interface

Since 90%+ Kenyans use WhatsApp, expose ALL i3 AI surfaces via WhatsApp: exam results, course content Q&A, policy bot, constituency feedback. One number. Zero app download needed. Uses existing n8n + WhatsApp Business API.

 Model Fallback + Cost Routing

LiteLLM already supports fallbacks. Implement cost-aware routing: try Qwen local first (free) → if overloaded, fail to GLM-4 local → if complex query, escalate to watsonx/KIMI API. 80% cost reduction vs. pure API usage.

 AI Analytics for Educators

Langfuse already traces all LLM calls. Build an "AI Usage Dashboard" for instructors: which students are using AI Lab most, what topics are they stuck on, which exam questions generated the most hint requests → curriculum intelligence.

 i3 Trust Layer (Ethics Shield)

Extend the Admissions Lobster Trap firewall to ALL interfaces. Add: hate speech detection (NCIC compliance), hallucination scoring, source confidence score, and an "AI Authenticity Certificate" shown on every AI-generated output — building user trust.

 Multi-Agent Collaboration

Deploy watsonx Orchestrate or LangGraph to chain agents: Research Agent → Draft Agent → Review Agent → Publish Agent. For PMaaS: Intel Agent → Narrative Agent → Approval Gate → Broadcast Agent. Fully automated with human gates at each approval step.

 Part 6 — Implementation Roadmap

PHASE	DURATION	DELIVERABLES	PRIORITY
Phase 1 Foundation	Weeks 1–4	Add Qwen2.5-7B + GLM-4-9B to LiteLLM. Clone admissions_agent.py as pmaas_policy_agent.py. Create i3-pmaas namespace. Add PMaaS Keycloak realm clients (campaign-manager, field-agent roles).	Critical
Phase 2 PMaaS Core	Weeks 5–10	Voice-clone TTS microservice (ElevenLabs). Telephony bridge (Twilio/Africa's Talking). WhatsApp bot via n8n. Sentiment heatmap engine. Manifesto RAG ingestion pipeline. War Room dashboard v1.	Critical
Phase 3 AI Interfaces	Weeks 8–14	i3 AI Chat frontend (chat.i3technologies.co.ke). Co-Worker sidebar (coworker.i3technologies.co.ke). Coding assistant (VS Code extension + JupyterHub %%ai magic). Voice input/output layer.	High
Phase 4 Advanced Models	Weeks 12–18	Add Qwen2.5-72B + Qwen2.5-Coder. KIMI K3 API integration. Add African language detection routing. Multi-agent orchestration. Agent marketplace (no-code builder). Cost-aware routing rules.	High

PHASE	DURATION	DELIVERABLES	PRIORITY
Phase 5 PMaaS Full Cycle	Months 4– 6	Deepfake monitor agent. Closing Argument Engine. Polling Day Dashboard. Governance AI bot. Constituency feedback system. Hansard monitor. Full War Room v2. PMaaS Enterprise tier live.	Medium



Part 7 — Summary: What to Do Next

- 1 **Create the i3-pmaas namespace** — mirror the pattern from other namespaces in `platform/namespaces/namespaces.yaml` . Add ResourceQuotas and LimitRanges for PMaaS workloads.
- 2 **Add Qwen2.5-7B and GLM-4-9B to the Ollama initContainer** in `vllm-deploy.yaml` — pull both models on boot alongside Granite 3.1. Register both in `litellm-deploy.yaml` `model_list`.
- 3 **Add MOONSHOT_API_KEY to OpenBao / .env.example** for KIMI K3 API access. Register kimi-k3 in LiteLLM config as Tier 1 model.
- 4 **Clone admissions_agent.py as pmaas_policy_agent.py** — swap the system prompt for the voter policy advisor persona. Ingest manifesto into ChromaDB collection `pmaas-manifesto` .
- 5 **Create a new Keycloak realm or client set for PMaaS** — roles: `campaign-manager` , `field-agent` , `analyst` , `candidate` . Route War Room to require `campaign-manager` role.
- 6 **Build the i3 AI Chat frontend** — Next.js app at `chat.i3technologies.co.ke` . WebSocket to FastAPI backend. Language auto-detect + model routing. Add to Argo CD app-of-apps.
- 7 **Plan voice-clone consent and legal review** — candidate consent forms, ElevenLabs API contract, DPA 2019 compliance sign-off before any voice data is processed.
- 8 **Design the Co-Worker sidebar** — React component embeddable in every i3 frontend. Context API for current-page awareness. Authenticated via Keycloak token.